

MySQL (Jointures)

- La jointure consiste à rechercher entre deux tables(ou plus) ayant un attribut commun (même type et même domaine de définition) tous les tuples (toutes les lignes) pour lesquels ces attributs ont la même valeur.
- MySQL prend en charge les types de jointures suivants :
 - INNER JOIN (Jointure interne)
 - LEFT JOIN (Jointure gauche)
 - RIGHT JOIN (Jointure droite)
 - CROSS JOIN (Jointure croisée)– Pour joindre des tables, On utilise la clause de jointure dans
 - l'instruction SELECT après la clause FROM.

Users

ID **Name**

1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

INNER JOIN

(or JOIN)



Likes

User ID **Like**

3	Stars
1	Climbing
1	Code
6	Rugby
4	Apples

JOIN

Name

Like

Maria	Stars
Patrik	Climbing
Patrik	Code
Darwin	Apples

```
SQL> SELECT users.name, likes.like
FROM users INNER JOIN likes
ON users.id = likes.user_id;
```

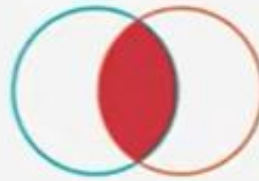
Users

ID Name

1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

INNER JOIN

(or JOIN)



Likes

User ID Like

3	Stars
1	Climbing
1	Code
6	Rugby
4	Apples

JOIN

Name	Like
Maria	Stars
Patrik	Climbing
Patrik	Code
Darwin	Apples

Users

ID Name

1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

LEFT JOIN



Likes

User ID Like

3	Stars
1	Climbing
1	Code
6	Rugby
4	Apples

JOIN

Name	Like
Patrik	Climbing
Patrik	Code
Albert	NULL
Maria	Stars
Darwin	Apples
Elizabeth	NULL

```
SQL> SELECT users.name, likes.like  
FROM users LEFT JOIN likes  
ON users.id = likes.user_id;
```

Users

ID	Name
1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

LEFT JOIN



JOIN

Name	Like
Patrik	Climbing
Patrik	Code
Albert	NULL
Maria	Stars
Darwin	Apples
Elizabeth	NULL

Likes

User ID	Like
3	Stars
1	Climbing
1	Code
6	Rugby
4	Apples

Users

ID Name

1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

RIGHT JOIN



Likes

User ID Like

3	Stars
1	Climbing
1	Code
6	Rugby
4	Apples

JOIN

Name Like

Maria	Stars
Patrik	Climbing
Patrik	Code
NULL	Rugby
Darwin	Apples

```
SQL> SELECT users.name, likes.like  
FROM users RIGHT JOIN likes  
ON users.id = likes.user_id;
```

Users

ID	Name
1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

RIGHT JOIN



Likes

User ID	Like
3	Stars
1	Climbing
1	Code
6	Rugby
4	Apples

JOIN

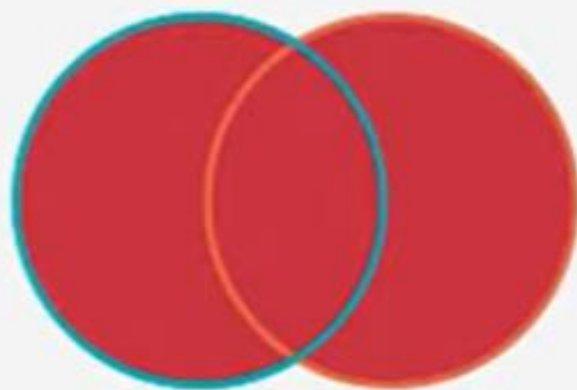
Name	Like
Maria	Stars
Patrik	Climbing
Patrik	Code
NULL	Rugby
Darwin	Apples

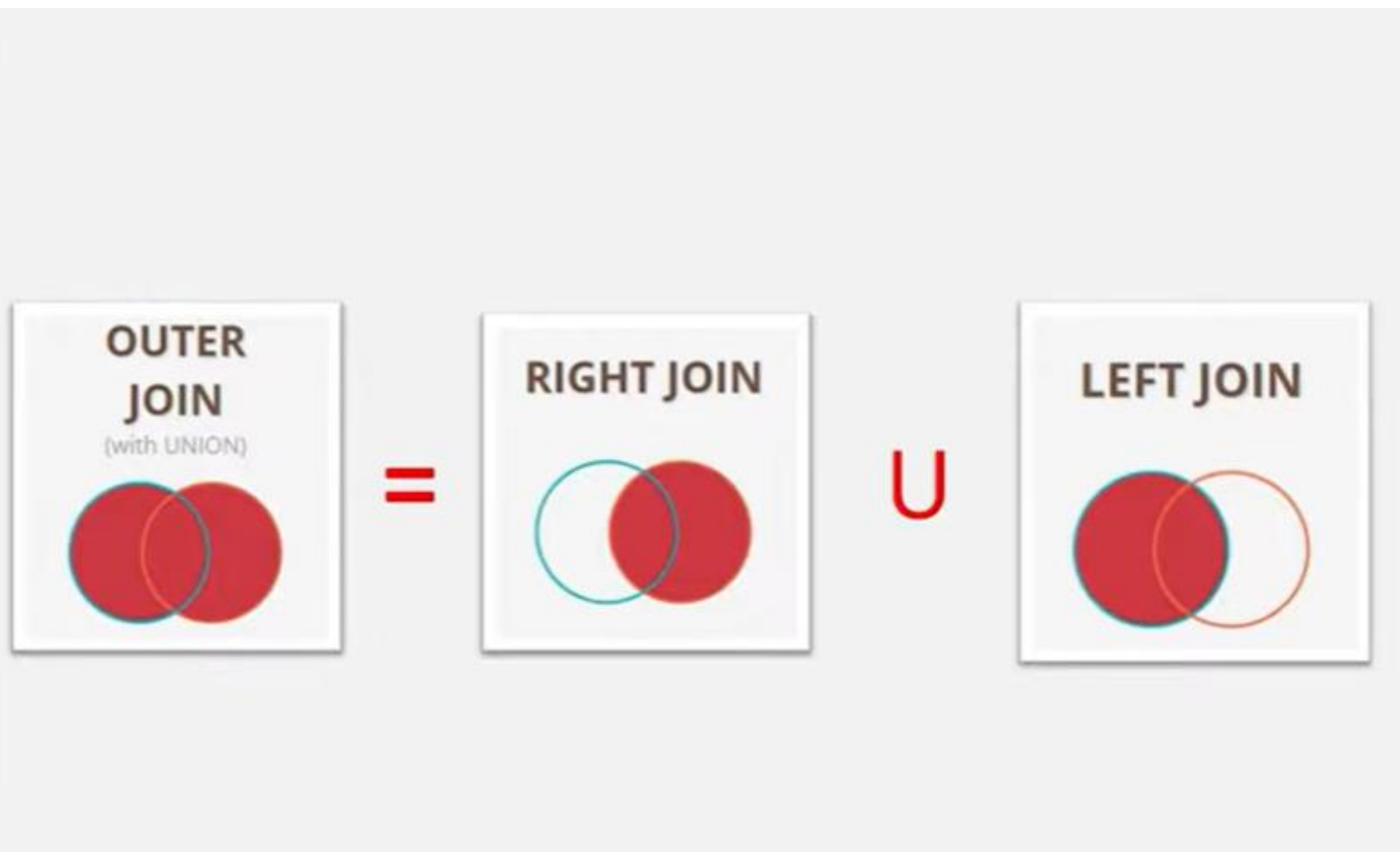
les types de jointures

OUTER

JOIN

(with UNION)





Users

ID Name

1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

OUTER JOIN

(with UNION)



Likes

User ID Like

3	Stars
1	Climbing
1	Code
6	Rugby
	Apples

JOIN

Name	Like
Patrik	Climbing
Patrik	Code
Albert	NULL
Maria	Stars
Darwin	Apples
Elizabeth	NULL
NULL	Rugby

```
SQL> SELECT users.name, likes.like
FROM users LEFT JOIN likes
ON users.id = likes.user_id
UNION
SELECT users.name, likes.like FROM
users RIGHT JOIN likes
ON users.id = likes.user_id;
```

Users

ID	Name
1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

Like

Stars
Climbing
Code
Rugby
Apples

JOIN

Name	Like
Patrik	Climbing
Patrik	Code
Albert	NULL
Maria	Stars
Darwin	Apples
Elizabeth	NULL
NULL	Rugby

```
SQL> SELECT users.name, likes.like  
FROM users FULL OUTER JOIN likes  
ON users.id = likes.user_id;
```

Users

ID	Name
1	Patrik
2	Albert
3	Maria
4	Darwin
5	Elizabeth

Likes

User ID	Like
3	Stars
	Climbing
	Code
	Rugby
	Apples

JOIN

Name	Like
Patrik	Climbing
Patrik	Code
Albert	NULL
Maria	Stars
Darwin	Apples
Elizabeth	NULL
NULL	Rugby